

Upcoming Dates

- **Today - Homework 12**
- **Sunday Night: Module 5 Video**
- **Thursday: 4-23: Module 5 Post Lab Worksheet**
- **Tuesday 4-28: HW 13 Due + Remote class work**

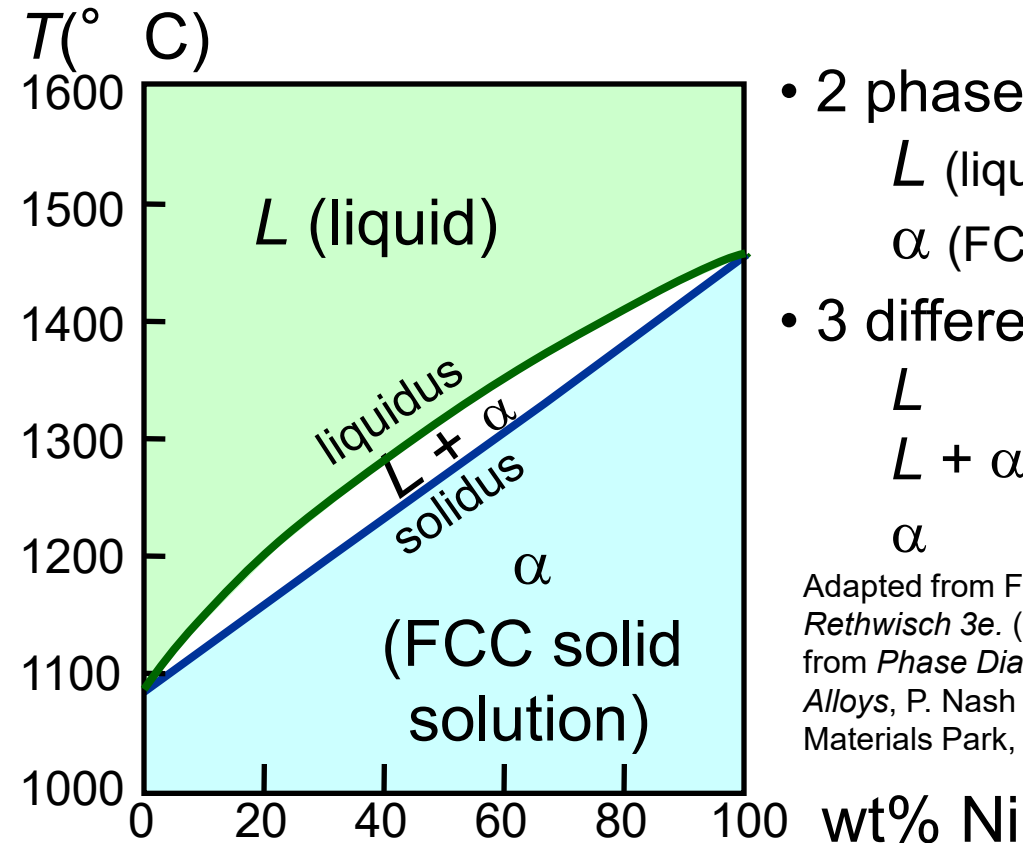
Classes Next Week

- **Tuesday – Honors Day**
 - **NO CLASSES**
- **Thursday – Capstone Design Showcase**
 - **NO Material Science Class**
 - **Remote Work will be assigned (including mini-lab)**
 - **You should also plan to attend the showcase**
 - **1:30 PM – 3:30 PM JLK atrium.**

Review: Phase Diagrams

- Indicate phases as a function of T , C , and P .
- For this course:
 - binary systems: just 2 components.
 - independent variables: T and C ($P = 1$ atm is almost always used).

Phase
Diagram
for Cu-Ni
system

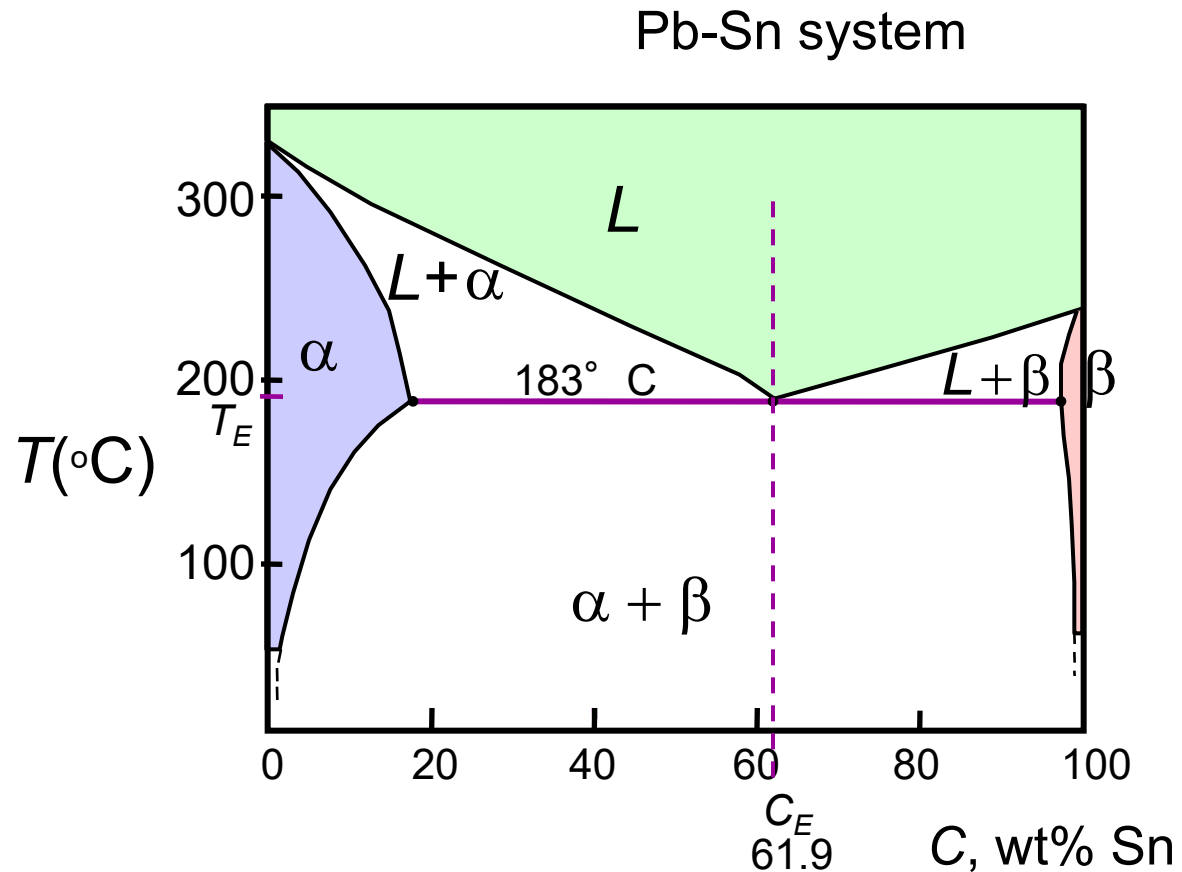


- 2 phases:
 - L (liquid)
 - α (FCC solid solution)
- 3 different phase fields:
 - L
 - $L + \alpha$
 - α

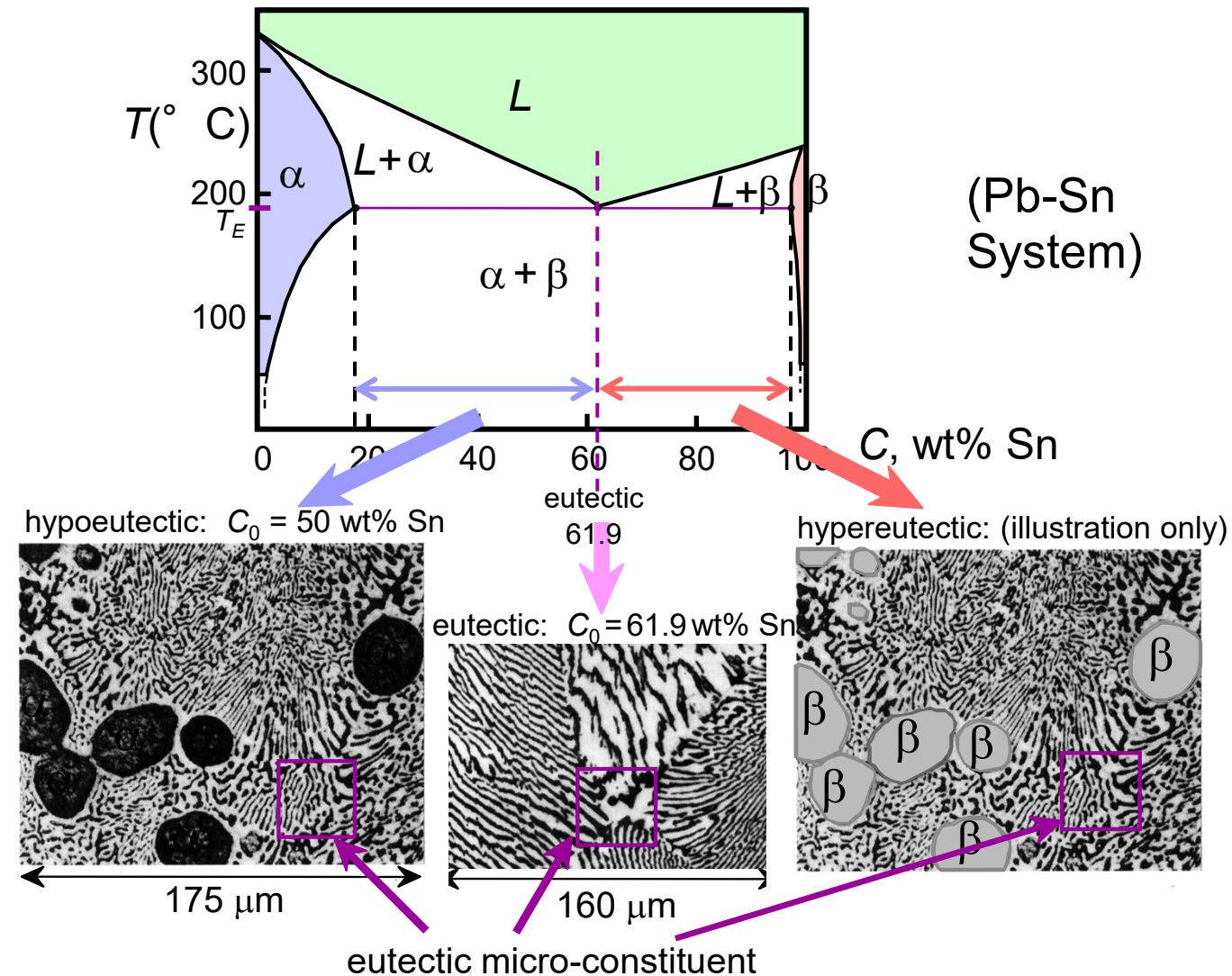
Adapted from Fig. 10.3(a), *Callister & Rethwisch 3e*. (Fig. 10.3(a) is adapted from *Phase Diagrams of Binary Nickel Alloys*, P. Nash (Ed.), ASM International, Materials Park, OH (1991).

Review: Eutectic Phase Diagram

T_E - eutectic temperature; C_E - eutectic composition

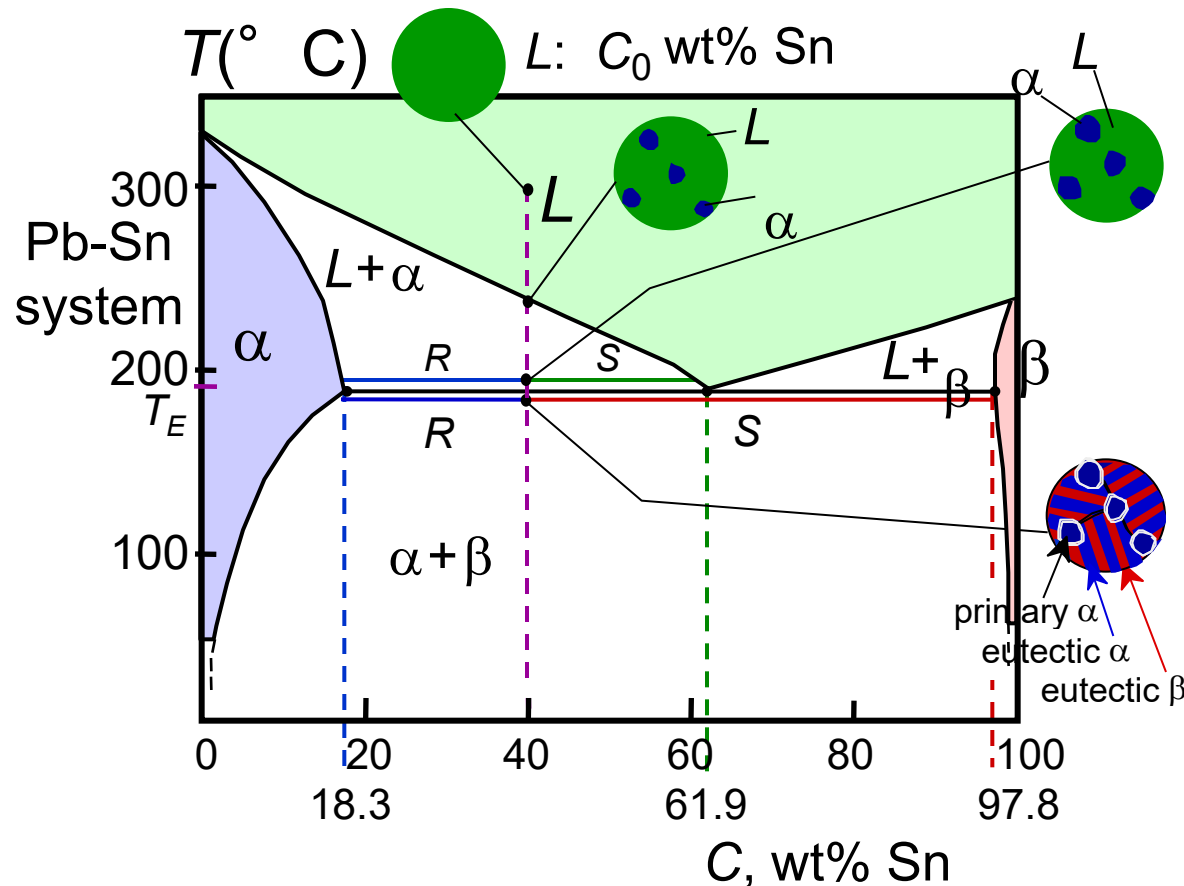


Hypoeutectic & Hypereutectic



Microstructural Developments in Eutectic Systems IV

- For alloys for which $18.3 \text{ wt\% Sn} < C_0 < 61.9 \text{ wt\% Sn}$
- Result: α phase particles and a eutectic microconstituent



- Just above T_E :

$$C_{\alpha} = 18.3 \text{ wt\% Sn}$$

$$C_L = 61.9 \text{ wt\% Sn}$$

$$W_{\alpha} =$$

$$W_L =$$

- Just below T_E :

$$C_{\alpha} = 18.3 \text{ wt\% Sn}$$

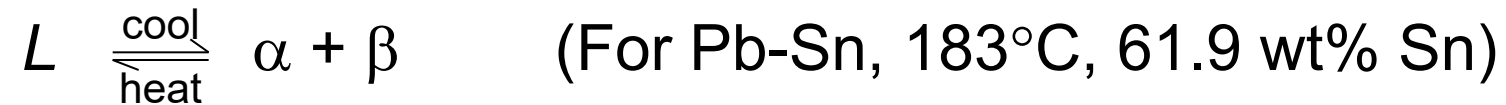
$$C_{\beta} = 97.8 \text{ wt\% Sn}$$

$$W_{\alpha} = \frac{S}{R + S} = 0.73$$

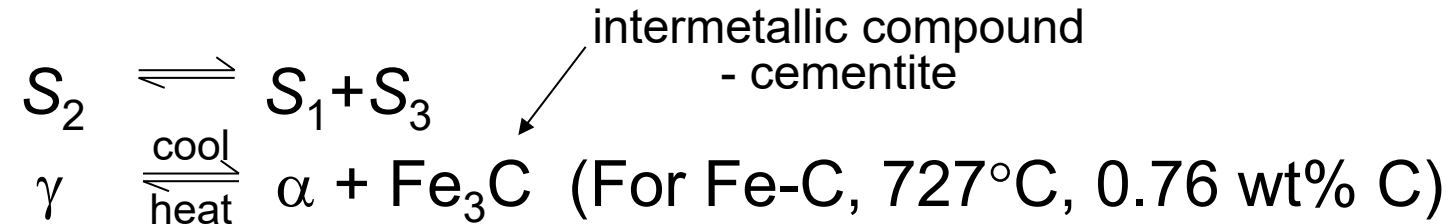
$$W_{\beta} = 0.27$$

Eutectic, Eutectoid, & Peritectic

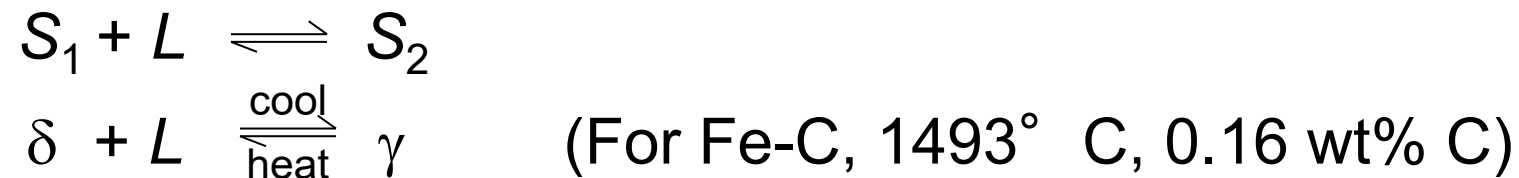
- Eutectic - liquid transforms to two solid phases



- Eutectoid – one solid phase transforms to two other solid phases

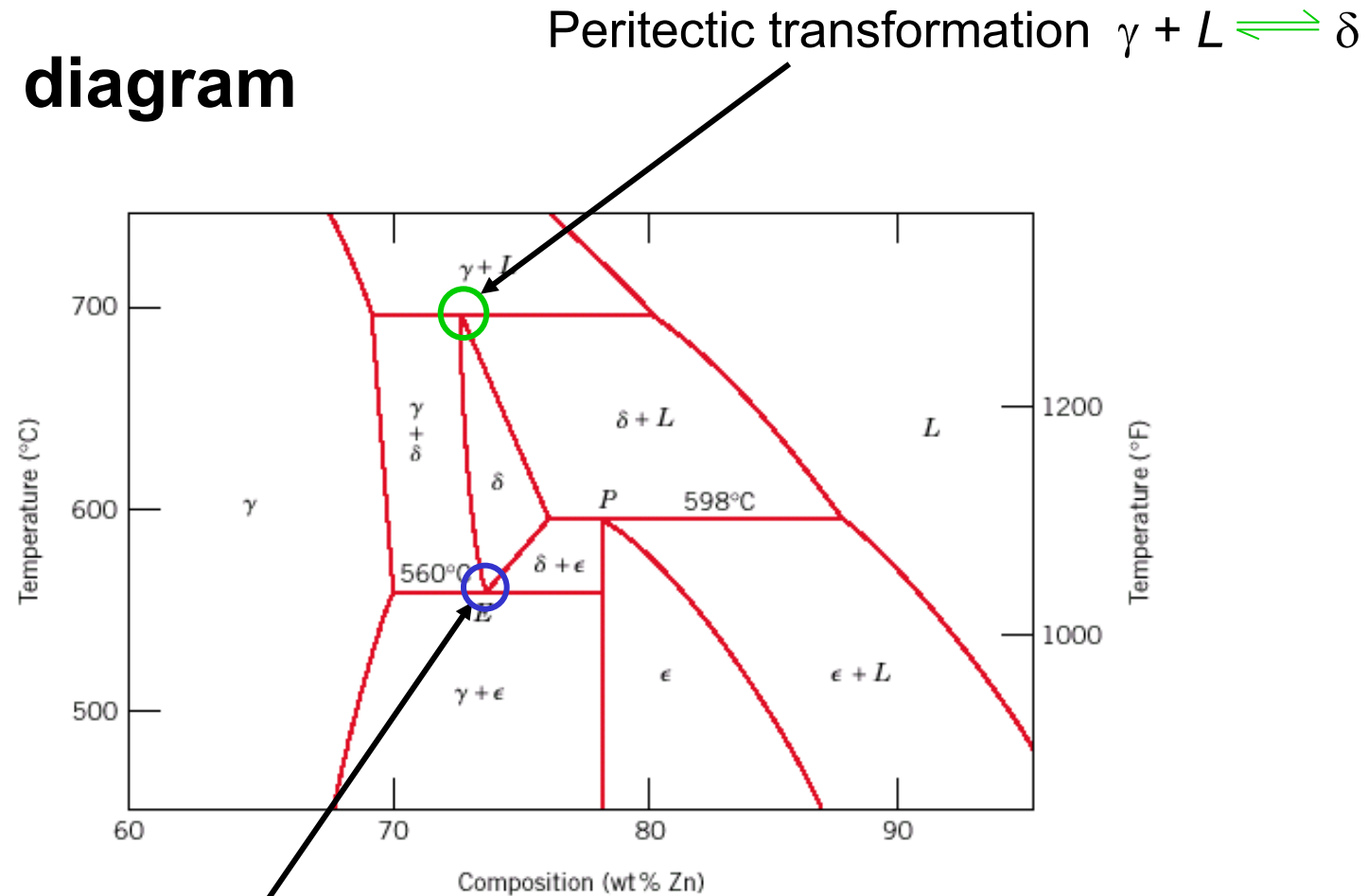


- Peritectic - liquid and one solid phase transform to a second solid phase



Eutectoid & Peritectic

Cu-Zn Phase diagram



Adapted from Fig. 10.21,
Callister & Rethwisch 3e.

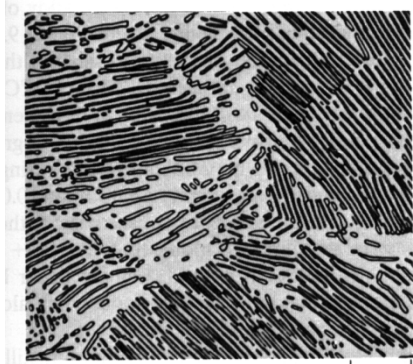
Eutectoid transformation $\delta \rightleftharpoons \gamma + \epsilon$

Iron-Carbon (Fe-C) Phase Diagram

- 2 important points

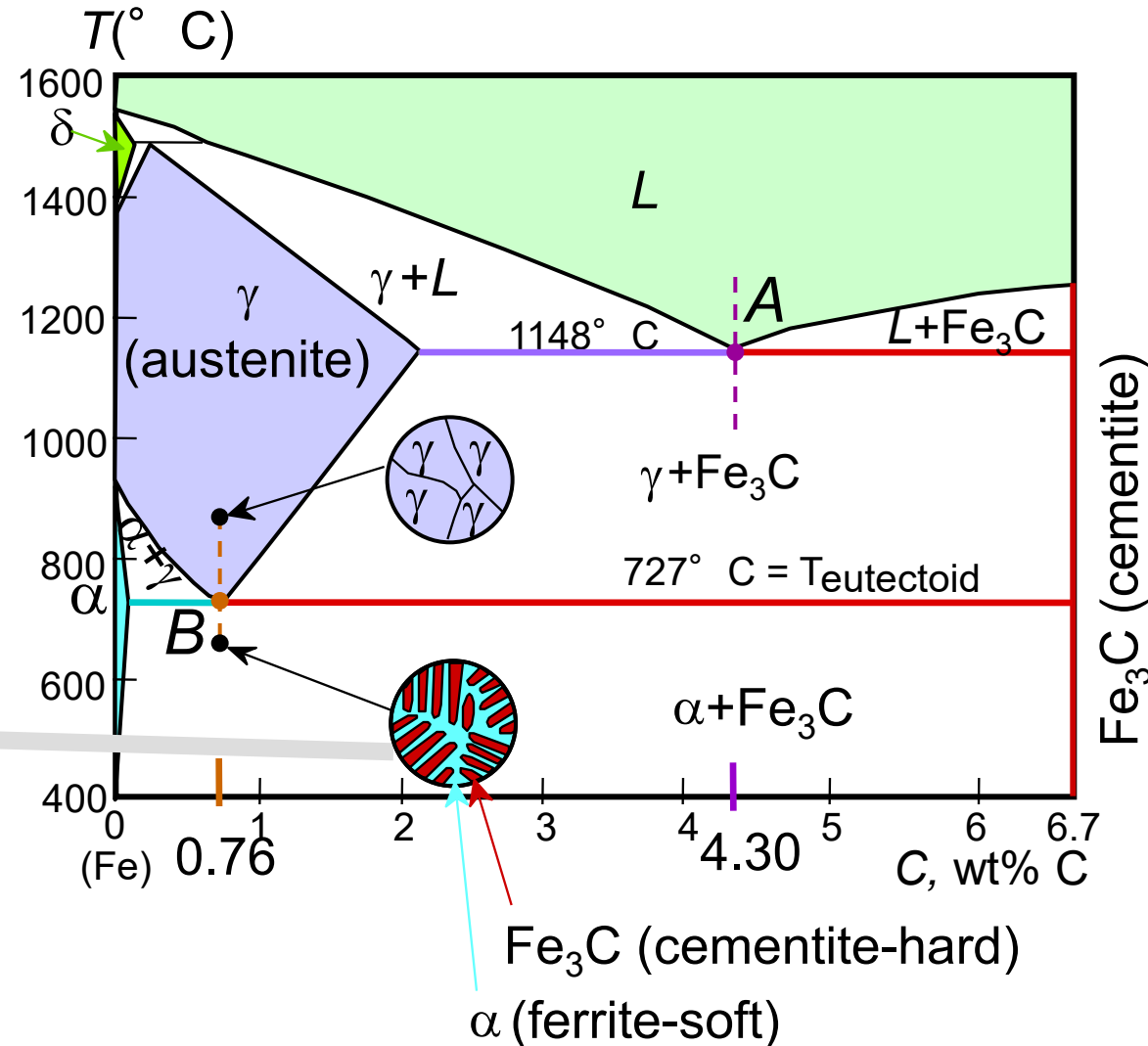
- Eutectic (A):
 $L \Rightarrow \gamma + \text{Fe}_3\text{C}$

- Eutectoid (B):
 $\gamma \Rightarrow \alpha + \text{Fe}_3\text{C}$

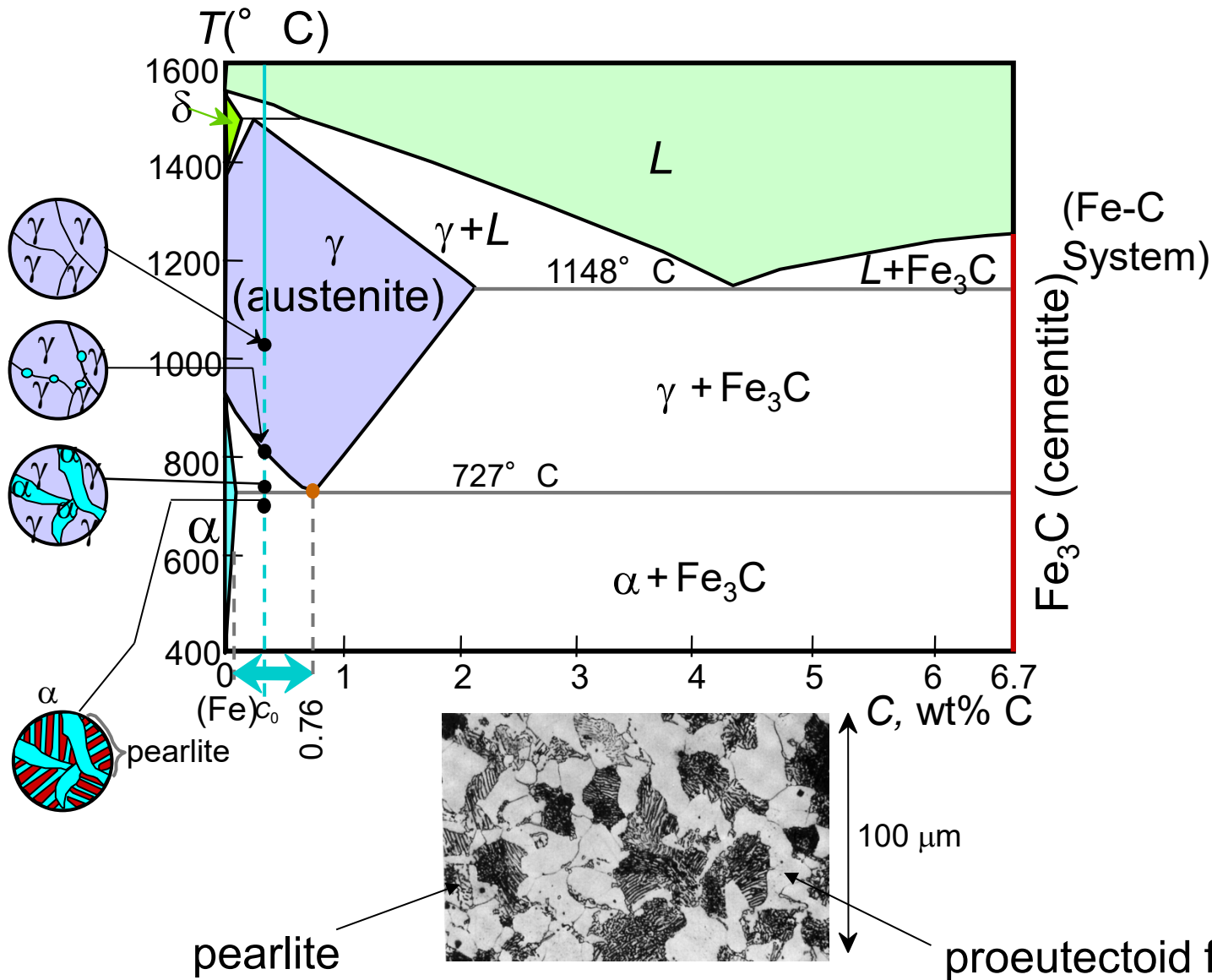


120 μm

Result: Pearlite = alternating layers of α and Fe_3C phases



Hypoeutectoid Steel

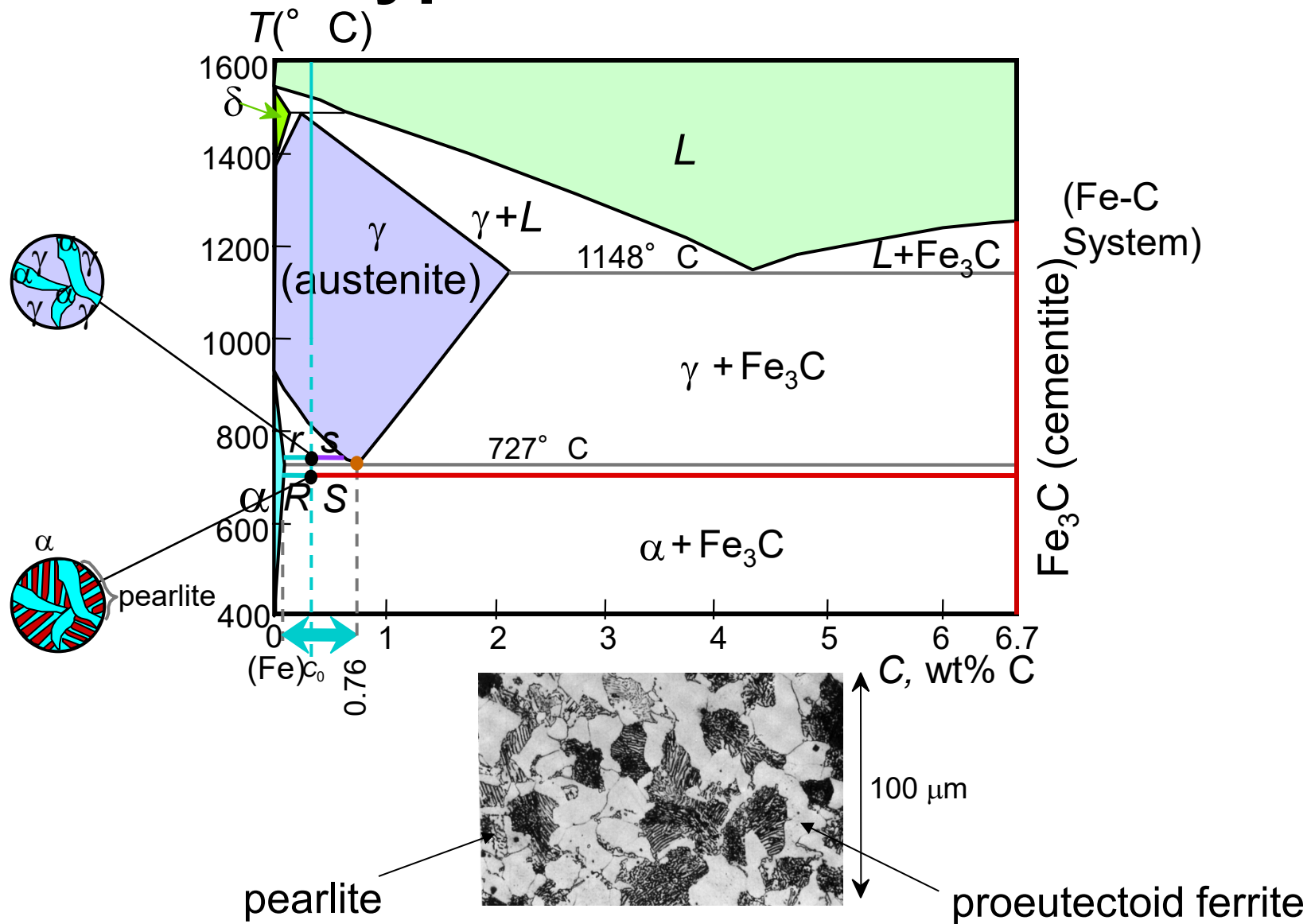


δ -ferrite, BCC: $1394\text{-}1538^{\circ}\text{C}$

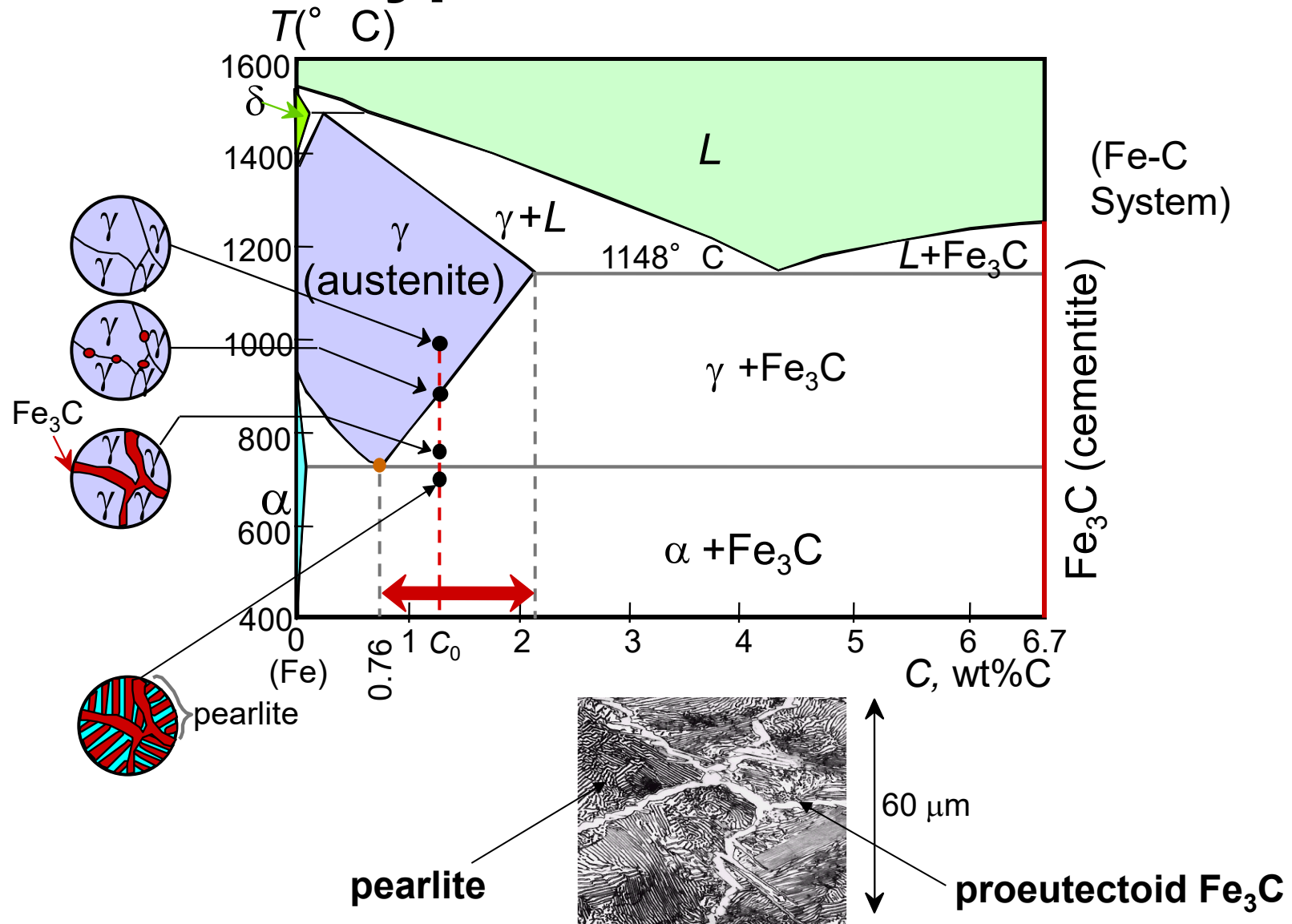
γ -austenite, FCC: $912\text{-}1394^{\circ}\text{C}$

α -ferrite, BCC: $<912^{\circ}\text{C}$

Hypoeutectoid Steel



Hypereutectoid Steel



Example Problem

For a 99.6 wt% Fe-0.40 wt% C steel at a temperature just below the eutectoid, determine the following:

- a) The compositions of Fe_3C and ferrite (α).
- b) The amount of cementite (in grams) that forms in 100 g of steel.

Solution to Example Problem

a) Using the *RS* tie line just below the eutectoid

$$C_{\alpha} = 0.022 \text{ wt\% C}$$

$$C_{\text{Fe}_3\text{C}} = 6.70 \text{ wt\% C}$$

b) Using the lever rule with the tie line shown

$$W_{\text{Fe}_3\text{C}} = \frac{R}{R+S} = \frac{C_0 - C_{\alpha}}{C_{\text{Fe}_3\text{C}} - C_{\alpha}}$$

$$= \frac{0.40 - 0.022}{6.70 - 0.022} = 0.057$$

Amount of Fe_3C in 100 g

$$= (100 \text{ g})W_{\text{Fe}_3\text{C}}$$

$$= (100 \text{ g})(0.057) = 5.7 \text{ g}$$

